

If Else Questions

1. Largest Among Three Numbers

Write a C program to input 3 numbers and print the largest number using if-else.

Program:

```
#include <stdio.h>

int main() {
    int a, b, c;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
    if (a >= b && a >= c)
        printf("Largest: %d\n", a);
    else if (b >= a && b >= c)
        printf("Largest: %d\n", b);
    else
        printf("Largest: %d\n", c);
    return 0;
}
```

Sample Output:

```
Enter three numbers: 45 78 23
Largest: 78
```

2. Leap Year Checker

Write a program to check whether a given year is a leap year or not.

□ Rules:

- Divisible by 4
- But not divisible by 100
- Unless divisible by 400

Program:

```
#include <stdio.h>
```

```

int main() {
    int year;
    printf("Enter year: ");
    scanf("%d", &year);
    if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
        printf("%d is a Leap Year.\n", year);
    else
        printf("%d is NOT a Leap Year.\n", year);
    return 0;
}

```

Sample Output:

```

Enter year: 2024
2024 is a Leap Year.

```

3. Electricity Bill Generator

Create a program to calculate electricity bill based on units:

Units	Rate
0-100	Rs.5/unit
101-200	Rs.7/unit
201+	Rs.10/unit

Add 18% GST at the end.

Program:

```

#include <stdio.h>

int main() {
    int units;
    float bill;
    printf("Enter units consumed: ");
    scanf("%d", &units);
    if (units <= 100)
        bill = units * 5;

```

```

        else if (units <= 200)
            bill = 100*5 + (units-100)*7;
        else
            bill = 100*5 + 100*7 + (units-200)*10;
        bill = bill + bill * 0.18;
        printf("Total Bill (with 18%% GST): Rs. %.2f\n", bill);
        return 0;
    }

```

Sample Output:

```

Enter units consumed: 150
Total Bill (with 18% GST): Rs. 885.00

```

4. Grade System

Input marks of 5 subjects and calculate percentage. Print:

- A Grade -> 90+
- B Grade -> 75+
- C Grade -> 50+
- Fail -> below 50 Also print:
- "Distinction"
- "Pass"
- "Fail"

Program:

```

#include <stdio.h>

int main() {
    float m1,m2,m3,m4,m5,total,per;
    printf("Enter marks of 5 subjects: ");
    scanf("%f %f %f %f %f",&m1,&m2,&m3,&m4,&m5);
    total = m1+m2+m3+m4+m5;
    per = total/5;
    printf("Percentage: %.2f%%\n", per);
    if(per >= 90)      printf("Grade: A\nDistinction\n");
    else if(per >= 75) printf("Grade: B\nPass\n");
    else if(per >= 50) printf("Grade: C\nPass\n");
}

```

```
        else                printf("Grade: Fail\nFail\n");
        return 0;
    }
}
```

Sample Output:

```
Enter marks of 5 subjects: 92 88 95 90 91
Percentage: 91.20%
Grade: A
Distinction
```

5. ATM Withdrawal System

Input:

- Account balance
- Withdrawal amount

Conditions:

- Minimum balance should remain Rs.1000
- Withdrawal should be multiple of 100
- Print success/failure message

Program:

```
#include <stdio.h>

int main() {
    float balance, amount;
    printf("Enter account balance: ");
    scanf("%f", &balance);
    printf("Enter withdrawal amount: ");
    scanf("%f", &amount);
    if((int)amount % 100 != 0)
        printf("Failed: Amount must be multiple of 100.\n");
    else if(balance - amount < 1000)
        printf("Failed: Minimum balance of Rs.1000 must remain.\n");
    else {
        balance -= amount;
        printf("Success! Withdrawn: Rs.%.0f | Remaining: Rs.%.2f\n",
            amount, balance);
    }
}
```

```
    }  
    return 0;  
}
```

Sample Output:

```
Enter account balance: 5000  
Enter withdrawal amount: 2000  
Success! Withdrawn: Rs.2000 | Remaining: Rs.3000.00
```

6. Login Authentication System

Create a simple login system. Correct credentials:

- Username: admin
- Password: 1234

Use nested if-else.

Program:

```
#include <stdio.h>  
#include <string.h>  
  
int main() {  
    char username[50], password[50];  
    printf("Enter Username: ");  
    scanf("%s", username);  
    printf("Enter Password: ");  
    scanf("%s", password);  
    if(strcmp(username, "admin") == 0) {  
        if(strcmp(password, "1234") == 0)  
            printf("Login Successful! Welcome, admin.\n");  
        else  
            printf("Login Failed: Incorrect Password.\n");  
    } else {  
        printf("Login Failed: Username not found.\n");  
    }  
    return 0;  
}
```

Sample Output:

Enter Username: admin

Enter Password: 1234

Login Successful! Welcome, admin.

7. Triangle Validator

Input 3 angles. Check:

- Whether triangle is valid
- Then print:
- Equilateral
- Isosceles
- Scalene

Program:

```
#include <stdio.h>

int main() {
    int a, b, c;
    printf("Enter three angles: ");
    scanf("%d %d %d", &a, &b, &c);
    if(a+b+c != 180 || a<=0 || b<=0 || c<=0)
        printf("Invalid Triangle!\n");
    else if(a==b && b==c)
        printf("Valid Triangle: Equilateral\n");
    else if(a==b || b==c || a==c)
        printf("Valid Triangle: Isosceles\n");
    else
        printf("Valid Triangle: Scalene\n");
    return 0;
}
```

Sample Output:

Enter three angles: 60 60 60

Valid Triangle: Equilateral

8. Employee Salary Bonus System

Input:

- Salary
- Years of experience
- Performance rating

Conditions:

- Experience > 5 years -> 20% bonus
- Rating A -> Extra 10%
- Rating B -> Extra 5%
- Otherwise no extra bonus

Print final salary.

Program:

```
#include <stdio.h>

int main() {
    float salary, bonus = 0;
    int exp;
    char rating;
    printf("Enter Salary, Years of Experience, Rating (A/B/C): ");
    scanf("%f %d %c", &salary, &exp, &rating);
    if(exp > 5) bonus += salary * 0.20;
    if(rating == 'A') bonus += salary * 0.10;
    else if(rating == 'B') bonus += salary * 0.05;
    printf("Bonus: Rs.%.2f\nFinal Salary: Rs.%.2f\n", bonus,
    salary+bonus);
    return 0;
}
```

Sample Output:

```
Enter Salary, Years of Experience, Rating (A/B/C): 50000 7 A
Bonus: Rs.15000.00
Final Salary: Rs.65000.00
```

9. E-Commerce Discount System

Create a shopping bill program. Conditions:

- Above Rs.5000 -> 20% discount
- Above Rs.2000 -> 10% discount

- VIP customer -> extra 5%

Add GST after discount.

Program:

```
#include <stdio.h>

int main() {
    float amount, discount = 0;
    char vip;
    printf("Enter bill amount: ");
    scanf("%f", &amount);
    printf("Are you a VIP customer? (Y/N): ");
    scanf(" %c", &vip);
    if(amount > 5000) discount = 0.20;
    else if(amount > 2000) discount = 0.10;
    if(vip == 'Y' || vip == 'y') discount += 0.05;
    amount = amount - amount * discount;
    amount = amount + amount * 0.18;
    printf("Discount: %.0f%%\nFinal Bill (with GST): Rs.%.2f\n",
    discount*100, amount);
    return 0;
}
```

Sample Output:

```
Enter bill amount: 6000
Are you a VIP customer? (Y/N): Y
Discount: 25%
Final Bill (with GST): Rs.5310.00
```

10. Online Voting System

Input:

- Age
- Citizenship (Indian or not)
- Criminal record (Yes/No)

Eligible only if:

- Age ≥ 18
- Indian citizen

- No criminal record

Program:

```
#include <stdio.h>
#include <string.h>
int main() {
    int age;
    char citizen[10], criminal[5];
    printf("Enter Age: ");
    scanf("%d", &age);
    printf("Are you an Indian citizen? (Yes/No): ");
    scanf("%s", citizen);
    printf("Do you have a criminal record? (Yes/No): ");
    scanf("%s", criminal);
    if(age >= 18 && strcmp(citizen,"Yes")==0 && strcmp(criminal,"No")==0)
        printf("You are ELIGIBLE to vote.\n");
    else
        printf("You are NOT ELIGIBLE to vote.\n");
    return 0;
}
```

Sample Output:

```
Enter Age: 22
Are you an Indian citizen? (Yes/No): Yes
Do you have a criminal record? (Yes/No): No
You are ELIGIBLE to vote.
```

11. Smart Traffic Signal System

Input vehicle speed. Conditions:

- Speed < 40 -> Safe
- 40-80 -> Normal
- 81-120 -> Warning
- Above 120 -> Challan + License Suspension

Program:

```

#include <stdio.h>

int main() {
    int speed;
    printf("Enter vehicle speed (km/h): ");
    scanf("%d", &speed);
    if(speed < 40)          printf("Status: SAFE\n");
    else if(speed <= 80)    printf("Status: NORMAL\n");
    else if(speed <= 120)   printf("Status: WARNING - Slow Down!\n");
    else                    printf("Status: CHALLAN issued + License
Suspended!\n");
    return 0;
}

```

Sample Output:

```

Enter vehicle speed (km/h): 130
Status: CHALLAN issued + License Suspended!

```

12. Income Tax Calculator

Calculate tax based on income slabs:

Income	Tax
Up to Rs.2.5L	0%
Rs.2.5L - Rs.5L	5%
Rs.5L - Rs.10L	20%
Above Rs.10L	30%

Add 4% cess.

Program:

```

#include <stdio.h>

int main() {
    float income, tax = 0;
    printf("Enter annual income (in Rs.): ");

```

```

scanf("%f", &income);
if(income <= 250000) tax = 0;
else if(income <= 500000) tax = (income-250000)*0.05;
else if(income <= 1000000) tax = 12500 + (income-500000)*0.20;
else tax = 112500 + (income-1000000)*0.30;
tax = tax + tax*0.04;
printf("Tax Payable (incl. 4% cess): Rs.%.2f\n", tax);
return 0;
}

```

Sample Output:

```

Enter annual income (in Rs.): 800000
Tax Payable (incl. 4% cess): Rs.62400.00

```

13. Railway Ticket Booking

Input:

- Age
- Gender
- Ticket class

Conditions:

- Senior citizen discount
- Female discount
- AC class extra charge

Print final fare.

Program:

```

#include <stdio.h>

int main() {
    int age; char gender, cls;
    float fare = 500.0;
    printf("Enter Age: "); scanf("%d",&age);
    printf("Gender (M/F): "); scanf(" %c",&gender);
    printf("Class (A=AC, G=General): "); scanf(" %c",&cls);
    if(age >= 60) fare *= 0.70;

```

```
    if(gender=='F' || gender=='f') fare *= 0.90;
    if(cls=='A' || cls=='a') fare += 300;
    printf("Final Fare: Rs.%.2f\n", fare);
    return 0;
}
```

Sample Output:

```
Enter Age: 65
Gender (M/F): M
Class (A=AC, G=General): A
Final Fare: Rs.650.00
```

14. Student Result Management System

Input marks of:

- C
- Java
- DBMS
- HTML
- React

Conditions:

- Any subject below 35 -> Fail
- Otherwise calculate:
- Percentage
- Grade
- Topper status (above 90%)

Use nested conditions.

Program:

```
#include <stdio.h>

int main() {
    float c, java, dbms, html, react, per;
    printf("Enter marks for C, Java, DBMS, HTML, React: ");
    scanf("%f %f %f %f %f", &c, &java, &dbms, &html, &react);
    if(c<35 || java<35 || dbms<35 || html<35 || react<35)
```

```
        printf("Result: FAIL (Subject below 35)\n");
    else {
        per = (c+java+dbms+html+react)/5;
        printf("Percentage: %.2f%%\n", per);
        if(per>=90) printf("Grade: A | Status: TOPPER\n");
        else if(per>=75) printf("Grade: B | Status: PASS\n");
        else if(per>=50) printf("Grade: C | Status: PASS\n");
        else printf("Grade: D | Status: PASS\n");
    }
    return 0;
}
```

Sample Output:

```
Enter marks for C, Java, DBMS, HTML, React: 92 88 95 91 90
Percentage: 91.20%
Grade: A | Status: TOPPER
```

15. Bank Loan Eligibility System

Input:

- Salary
- CIBIL score
- Existing loan
- Age

Conditions:

- Salary > Rs.25,000
- CIBIL > 750
- Existing loan < Rs.5L
- Age between 21-60

Print:

- Eligible
- Not Eligible
- Premium Customer Loan Offer

Program:

```
#include <stdio.h>

int main() {
    float salary, existingLoan;
    int cibil, age;
    printf("Enter Salary, CIBIL Score, Existing Loan (Rs.), Age: ");
    scanf("%f %d %f %d",&salary,&cibil,&existingLoan,&age);
    if(salary>25000 && cibil>750 && existingLoan<500000 && age>=21 &&
age<=60) {
        if(salary>75000 && cibil>800)
            printf("Premium Customer Loan Offer Approved!\n");
        else
            printf("Eligible for Loan.\n");
    } else {
        printf("Not Eligible for Loan.\n");
    }
    return 0;
}
```

Sample Output:

```
Enter Salary, CIBIL Score, Existing Loan (Rs.), Age: 80000 820 200000 35
Premium Customer Loan Offer Approved!
```

For Loop Questions

1. Factorial of Number

Calculate factorial using for loop. Example: $5! = 120$

Program:

```
#include <stdio.h>

int main() {
    int n, i;
    long long fact = 1;
    printf("Enter a number: ");
    scanf("%d", &n);
    for(i = 1; i <= n; i++)
        fact *= i;
    printf("%d! = %lld\n", n, fact);
    return 0;
}
```

Sample Output:

```
Enter a number: 5
5! = 120
```

2. Reverse Table

Print multiplication table in reverse order. Example:

$5 \times 10 = 50$

$5 \times 9 = 45$

Program:

```
#include <stdio.h>

int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    for(int i = 10; i >= 1; i--)
        printf("%d x %d = %d\n", n, i, n*i);
}
```

```
        return 0;
    }
```

Sample Output:

```
Enter a number: 5
5 x 10 = 50
5 x 9  = 45
5 x 8  = 40
...
5 x 1  = 5
```

3. Count Digits

Input a number and count total digits using loop.

Program:

```
#include <stdio.h>

int main() {
    long long n;
    int count = 0;
    printf("Enter a number: ");
    scanf("%lld", &n);
    if(n == 0) count = 1;
    if(n < 0) n = -n;
    for(; n != 0; n /= 10) count++;
    printf("Total digits: %d\n", count);
    return 0;
}
```

Sample Output:

```
Enter a number: 123456
Total digits: 6
```

4. Sum of Digits

Find sum of digits of a number. Example: 1234 -> 10

Program:

```
#include <stdio.h>

int main() {
    long long n;
    int sum = 0;
    printf("Enter a number: ");
    scanf("%lld", &n);
    if(n < 0) n = -n;
    for(; n != 0; n /= 10)
        sum += n % 10;
    printf("Sum of digits: %d\n", sum);
    return 0;
}
```

Sample Output:

```
Enter a number: 1234
Sum of digits: 10
```

5. Reverse Number

Reverse a number using loop. Example: 1234 -> 4321

Program:

```
#include <stdio.h>

int main() {
    long long n, rev = 0;
    printf("Enter a number: ");
    scanf("%lld", &n);
    for(; n != 0; n /= 10)
        rev = rev * 10 + n % 10;
    printf("Reversed: %lld\n", rev);
    return 0;
}
```

Sample Output:

Enter a number: 1234

Reversed: 4321

6. Strong Number

Check whether number is Strong Number. Example: 145 -> Strong Number

Program:

```
#include <stdio.h>

int main() {
    int n, temp, digit, sum = 0, fact;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;
    while(temp) {
        digit = temp % 10;
        fact = 1;
        for(int i = 1; i <= digit; i++) fact *= i;
        sum += fact;
        temp /= 10;
    }
    if(sum == n) printf("%d is a Strong Number.\n", n);
    else printf("%d is NOT a Strong Number.\n", n);
    return 0;
}
```

Sample Output:

Enter a number: 145

145 is a Strong Number.

7. Find HCF and LCM

Find HCF and LCM of two numbers using loops.

Program:

```
#include <stdio.h>

int main() {
    int a, b, hcf, lcm, temp;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    int x = a, y = b;
    while(y) { temp = y; y = x % y; x = temp; }
    hcf = x;
    lcm = (a * b) / hcf;
    printf("HCF = %d\nLCM = %d\n", hcf, lcm);
    return 0;
}
```

Sample Output:

```
Enter two numbers: 12 18
HCF = 6
LCM = 36
```

8. Perfect Number

Check whether a number is perfect or not. Example: $6 = 1 + 2 + 3$

Program:

```
#include <stdio.h>

int main() {
    int n, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    for(int i = 1; i < n; i++)
        if(n % i == 0) sum += i;
    if(sum == n) printf("%d is a Perfect Number.\n", n);
    else        printf("%d is NOT a Perfect Number.\n", n);
}
```

```
    return 0;
}
```

Sample Output:

```
Enter a number: 6
6 is a Perfect Number.
```

9. Decimal to Binary Conversion

Convert decimal number to binary using loop.

Program:

```
#include <stdio.h>

int main() {
    int n, binary[32], i = 0;
    printf("Enter decimal number: ");
    scanf("%d", &n);
    if(n == 0) { printf("Binary: 0\n"); return 0; }
    while(n > 0) { binary[i++] = n % 2; n /= 2; }
    printf("Binary: ");
    for(int j = i-1; j >= 0; j--) printf("%d", binary[j]);
    printf("\n");
    return 0;
}
```

Sample Output:

```
Enter decimal number: 10
Binary: 1010
```

10. Menu Driven Calculator

Create calculator using:

- for
- switch
- loops

Features:

- Addition
- Subtraction
- Multiplication
- Division
- Continue option

Program:

```
#include <stdio.h>

int main() {
    int choice;
    float a, b, result;
    for(;;) {
        printf("\n1.Add 2.Sub 3.Mul 4.Div 5.Exit\nChoice: ");
        scanf("%d", &choice);
        if(choice == 5) break;
        printf("Enter two numbers: ");
        scanf("%f %f", &a, &b);
        switch(choice) {
            case 1: result=a+b; printf("Result: %.2f\n",result); break;
            case 2: result=a-b; printf("Result: %.2f\n",result); break;
            case 3: result=a*b; printf("Result: %.2f\n",result); break;
            case 4:
                if(b==0) printf("Error: Division by zero!\n");
                else { result=a/b; printf("Result: %.2f\n",result); }
                break;
            default: printf("Invalid choice!\n");
        }
    }
    return 0;
}
```

Sample Output:

```
1.Add 2.Sub 3.Mul 4.Div 5.Exit
Choice: 1
Enter two numbers: 15 25
```

Result: 40.00

11. Student Result System

Input marks of 5 students using loop and print:

- Total
- Percentage
- Grade

Program:

```
#include <stdio.h>

int main() {
    float marks[5], total, per;
    for(int i = 0; i < 5; i++) {
        total = 0;
        printf("\nStudent %d - Enter marks of 5 subjects: ", i+1);
        for(int j = 0; j < 5; j++) { scanf("%f",&marks[j]); total +=
marks[j]; }
        per = total / 5;
        printf("Total: %.0f | Percentage: %.2f%% | Grade: ", total, per);
        if(per>=90) printf("A\n");
        else if(per>=75) printf("B\n");
        else if(per>=50) printf("C\n");
        else printf("Fail\n");
    }
    return 0;
}
```

Sample Output:

```
Student 1 - Enter marks of 5 subjects: 80 75 90 85 88
Total: 418 | Percentage: 83.60% | Grade: B
```

12. Employee Salary System

Input salary of multiple employees and calculate:

- Bonus

- Tax
- Final Salary

Using loops + conditions.

Program:

```
#include <stdio.h>

int main() {
    int n;
    float salary, bonus, tax, final;
    printf("Enter number of employees: ");
    scanf("%d", &n);
    for(int i = 0; i < n; i++) {
        printf("\nEmployee %d - Enter Salary: ", i+1);
        scanf("%f", &salary);
        bonus = (salary > 50000) ? salary*0.20 : salary*0.10;
        tax    = (salary > 50000) ? salary*0.30 : salary*0.10;
        final = salary + bonus - tax;
        printf("Bonus: %.2f | Tax: %.2f | Final: %.2f\n", bonus, tax,
final);
    }
    return 0;
}
```

Sample Output:

```
Enter number of employees: 2
Employee 1 - Enter Salary: 60000
Bonus: 12000.00 | Tax: 18000.00 | Final: 54000.00
Employee 2 - Enter Salary: 30000
Bonus: 3000.00 | Tax: 3000.00 | Final: 30000.00
```

13. Online Exam Result System

Input marks of 5 subjects for multiple students. Using loop:

- Calculate:
- Total
- Percentage

- Grade
- Print Pass/Fail

Program:

```
#include <stdio.h>

int main() {
    int n; float marks[5], total, per; int pass;
    printf("Enter number of students: ");
    scanf("%d", &n);
    for(int i = 0; i < n; i++) {
        total = 0; pass = 1;
        printf("\nStudent %d - Enter marks for 5 subjects: ", i+1);
        for(int j = 0; j < 5; j++) {
            scanf("%f", &marks[j]);
            total += marks[j];
            if(marks[j] < 35) pass = 0;
        }
        per = total / 5;
        printf("Total: %.0f | Percentage: %.2f%%\n", total, per);
        if(!pass) { printf("Result: FAIL\n"); continue; }
        if(per>=90) printf("Grade: A | PASS\n");
        else if(per>=75) printf("Grade: B | PASS\n");
        else if(per>=50) printf("Grade: C | PASS\n");
        else printf("Grade: D | PASS\n");
    }
    return 0;
}
```

Sample Output:

```
Enter number of students: 1
Student 1 - Enter marks for 5 subjects: 88 92 78 95 85
Total: 438 | Percentage: 87.60%
Grade: B | PASS
```

14. Mobile Recharge System

A shopkeeper recharges multiple customers. Recharge Plans:

- Rs.199
- Rs.399
- Rs.599

Using loop:

- Input customer recharge
- Calculate total earnings of shopkeeper

Program:

```
#include <stdio.h>

int main() {
    int n, plan;
    float total = 0;
    printf("Enter number of customers: ");
    scanf("%d", &n);
    for(int i = 0; i < n; i++) {
        printf("Customer %d - Plan (1=Rs.199, 2=Rs.399, 3=Rs.599): ",
i+1);

        scanf("%d", &plan);
        if(plan==1) total += 199;
        else if(plan==2) total += 399;
        else if(plan==3) total += 599;
        else printf("Invalid plan!\n");
    }
    printf("Total Earnings: Rs.%.0f\n", total);
    return 0;
}
```

Sample Output:

```
Enter number of customers: 3
Customer 1 - Plan (1=Rs.199, 2=Rs.399, 3=Rs.599): 1
Customer 2 - Plan (1=Rs.199, 2=Rs.399, 3=Rs.599): 3
Customer 3 - Plan (1=Rs.199, 2=Rs.399, 3=Rs.599): 2
Total Earnings: Rs.1197
```

15. Hospital Patient Temperature Monitor

A hospital records temperature of patients. Using loop:

- Input temperature of 20 patients
- Count:
- Normal patients
- Fever patients (>100 F)

Program:

```
#include <stdio.h>

int main() {
    float temp;
    int normal = 0, fever = 0;
    for(int i = 1; i <= 20; i++) {
        printf("Enter temperature of patient %d (F): ", i);
        scanf("%f", &temp);
        if(temp > 100.0) fever++;
        else normal++;
    }
    printf("\n--- Report ---\n");
    printf("Normal Patients: %d\n", normal);
    printf("Fever Patients:  %d\n", fever);
    return 0;
}
```

Sample Output:

```
Enter temperature of patient 1 (F): 98.6
Enter temperature of patient 2 (F): 102.1
...
--- Report ---
Normal Patients: 14
Fever Patients:  6
```